

Many-body physics: introduction, notations and definitions

FYS4480/FYS9480 – Week 34, August 20–21, 2026

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Thursday August 20: linear algebra reminder, Dirac notation, Hamiltonians and basis sets

Friday August 21: fermions, bosons, Slater determinants and expectation values

Reading: these slides, chapters 1–2 of the lecture notes, Szabo and Ostlund chapters 1–2.

Code: `WeeklySlides/week34.ipynb`

Plan for week 34

Thursday August 20, 2×45 min

- Introduction to many-body physics and the plan for the semester
- Linear algebra reminder: vectors, matrices, inner and outer products
- Dirac notation, projection operators, density matrices, Pauli matrices
- Orthonormal bases, unitary transformations, tensor products, spectral decomposition
- Hamiltonians and basis sets

Friday August 21, 2×45 min

- Fermions and bosons, permutation symmetry and the Pauli principle
- Determinants and Slater determinants; changing the single-particle basis
- The variational principle and quantum mechanical expectation values
- The energy functional: direct and exchange terms, preview of Hartree-Fock
- Exercise session: exercise set for week 34

How these slides work

- Roughly one slide per minute; derivations are written out in full so that the slides can be read afterwards.
- Every lecture contains a few **pause to think** questions:

Pause to think #1

(about 2 min – discuss with your neighbour)

Why is the inner product $\langle x|x \rangle$ always a real number, even for a complex vector $|x\rangle$?

- Discuss with your neighbour for a couple of minutes, then we compare notes. The answer is always on the slide that follows.
- The companion notebook `week34.ipynb` runs every numerical example shown here.

Answer, and the rule we will use all semester

Answer to pause #1

$\langle x|x \rangle = \sum_i x_i^* x_i = \sum_i |x_i|^2 \geq 0$. Each term is a modulus squared, hence real and non-negative. For two *different* vectors $\langle y|x \rangle$ is complex in general, and $\langle x|y \rangle^* = \langle y|x \rangle$.

This is the first of many places where a property of ordinary vectors becomes a statement about quantum states: the norm of a state is real, and a probability is $|\langle \phi|\psi \rangle|^2$.

Thursday, lecture 1: Introduction and linear algebra reminder

Introduction to the course

These lectures aim at giving an introduction to the quantum mechanics of many-body systems and the methods relevant for many-body problems in such diverse areas as atomic, molecular, solid-state and nuclear physics, chemistry and materials science.

A theoretical understanding of the behaviour of quantum-mechanical many-body systems – systems containing many interacting particles – is a considerable challenge in that, normally, **no exact solution can be found**. Instead, reliable methods are needed for approximate but accurate simulations of such systems.

One sentence summary of the semester

The many-body problem is linear algebra in a space that is exponentially large; every method we study is a strategy for not having to look at all of it.

Content of the lecture notes, part 1

Introduction: definitions, examples and the mathematics of many-body functions

- Slater determinants, permutation operators, linear algebra reminder (determinants, vector and matrix algebra, tensor products, representations, unitary transformations)
- Ansätze for wave functions

Second quantization for bosons and fermions

- Commutation rules, creation and annihilation operators; Wick's theorem and its generalisation; the particle-hole picture
- Interaction, Schrödinger and Heisenberg pictures; time-dependent Wick theorem, Gell-Mann and Low, adiabatic switching
- One-, two- and three-body parts of Hamiltonians; Jordan-Wigner transformation; Baker-Campbell-Hausdorff and Suzuki-Trotter

Content of the lecture notes, part 2

Full configuration interaction (FCI)

- Basics of FCI, particle-hole rewriting
- Slater determinants, similarity transformations, eigenvalue algorithms
- Eigenvector continuation; diagrams

Mean-field theories

- Hartree-Fock in coordinate space and in second quantization
- Thouless' theorem, links to DFT, the electron gas

Many-body perturbation theory

- Time-dependent and time-independent; Brillouin-Wigner and Rayleigh-Schrödinger
- Diagrams; the linked-diagram theorem

Coupled cluster theory, and more

- Singles and doubles, standard and unitary; CCD versus FCI and MBPT
- Green's functions, time-dependent theory, quantum computing (VQE), applications (electron gas, Lipkin and pairing models, nuclear matter)

Textbooks and teaching mode

Recommended texts

- Szabo and Ostlund, *Modern Quantum Chemistry* (this week: chapters 1 and 2)
- Shavitt and Bartlett, *Many-body Methods in Chemistry and Physics*
- Weekly reading assignments and lecture notes (with programs and exercises) are sent before each week.

Teaching mode (hybrid: on-site and online lectures, on-site exercise sessions)

- Four lectures per week, fall semester, 10 ECTS; lectures are recorded.
- Two hours of exercise sessions for work on projects and exercises.
- Two graded projects, 25 % each; final oral/written exam, 50 %.
- Offered as FYS4480 (MSc) and FYS9480 (PhD). Weekly e-mails summarise the activities.

Notations and definitions

- Vectors, matrices and higher-order tensors are always **boldfaced**: vectors by lower-case letters ($\mathbf{x}, \mathbf{v}$), matrices and tensors by upper-case letters ($\mathbf{A}, \mathbf{U}$).
- Unless otherwise stated the elements v_i of a vector are real: $\mathbf{x} \in \mathbb{R}^n$; complex vectors are $\mathbf{x} \in \mathbb{C}^n$; a real $n \times n$ matrix is $\mathbf{A} \in \mathbb{R}^{n \times n}$.
- **Index counting starts at zero**, a_{00} to $a_{n-1,n-1}$, so that formulae and Python code carry the same indices.

Shorthand symbols

$\forall$ for all, $\implies$ implies, $\equiv$ equivalent / defined as; $\mathbb{R}, \mathbb{I}, \mathbb{C}$ real, integer and complex numbers.

Vectors

A vector $\mathbf{x}$ with n components, its transpose and (for complex vectors) its Hermitian conjugate:

$$\mathbf{x} = \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_{n-1} \end{bmatrix}, \quad \mathbf{x}^T = [x_0 \quad x_1 \quad \cdots \quad x_{n-1}], \quad \mathbf{x}^\dagger = [x_0^* \quad x_1^* \quad \cdots \quad x_{n-1}^*].$$

The inner (dot) product is a scalar,

$$\mathbf{x}^T \mathbf{x} = \sum_{i=0}^{n-1} x_i x_i = x_0^2 + x_1^2 + \cdots + x_{n-1}^2, \quad \mathbf{x}^\dagger \mathbf{x} = \sum_{i=0}^{n-1} x_i^* x_i = \|\mathbf{x}\|^2.$$

Vector operations we will meet all the time:

addition/subtraction: $\mathbf{x} = \mathbf{y} \pm \mathbf{z} \implies x_i = y_i \pm z_i,$

scalar multiplication: $\mathbf{x} = \gamma \mathbf{y} \implies x_i = \gamma y_i,$

Hadamard (element-wise) product: $\mathbf{x} = \mathbf{y} \circ \mathbf{z} \implies x_i = y_i z_i.$

Outer products

The outer product of x (length n) and y (length m) is the $n \times m$ matrix

$$xy^T = \begin{bmatrix} x_0 y_0 & x_0 y_1 & \cdots & x_0 y_{m-1} \\ x_1 y_0 & x_1 y_1 & \cdots & x_1 y_{m-1} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n-1} y_0 & x_{n-1} y_1 & \cdots & x_{n-1} y_{m-1} \end{bmatrix}, \quad (xy^T)_{ij} = x_i y_j,$$

and for complex vectors $(xy^\dagger)_{ij} = x_i y_j^*$.

The outer product also fixes our basic matrix layout: row index first, column index second. In quantum mechanics outer products become **projection operators** and **density matrices** – we get there in a few slides.

Matrices

A general $n \times n$ matrix, and the same matrix written in terms of its column vectors a_i :

$$A = \begin{bmatrix} a_{00} & a_{01} & \cdots & a_{0,n-1} \\ a_{10} & a_{11} & \cdots & a_{1,n-1} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n-1,0} & a_{n-1,1} & \cdots & a_{n-1,n-1} \end{bmatrix} = [a_0 \quad a_1 \quad \cdots \quad a_{n-1}].$$

scalar multiplication: $A = \gamma B \implies a_{ij} = \gamma b_{ij},$

addition/subtraction: $A = B \pm C \implies a_{ij} = b_{ij} \pm c_{ij},$

matrix-vector product: $y = Ax \implies y_i = \sum_{j=0}^{n-1} a_{ij} x_j,$

matrix-matrix product: $A = BC \implies a_{ij} = \sum_{k=0}^{n-1} b_{ik} c_{kj},$

transpose: $A = B^T \implies a_{ij} = b_{ji}.$

Inverse, Hermitian conjugate and nonsingularity

- The inverse of a square matrix (if it exists) is defined by $A^{-1}A = I$, with I the unit matrix.
- The Hermitian conjugate is obtained by transposing and complex-conjugating every element: $(A^\dagger)_{ij} = a_{ji}^*$. We will mostly deal with Hermitian (and square) matrices, so "transpose" and "adjoint" are often used loosely.

Equivalent statements for an $n \times n$ matrix A

1. The inverse of A exists (A is nonsingular).
2. $Ax = 0$ implies $x = 0$.
3. The rows of A form a basis of $\mathbb{C}^n$.
4. The columns of A form a basis of $\mathbb{C}^n$.
5. A is a product of elementary matrices. (Can you name one?)
6. 0 is not an eigenvalue of A .

Selected matrix types

Relation	Name	Matrix elements
$A = A^T$	symmetric	$a_{ij} = a_{ji}$
$A = (A^T)^{-1}$	real orthogonal	$\sum_k a_{ik} a_{jk} = \sum_k a_{ki} a_{kj} = \delta_{ij}$
$A = A^*$	real matrix	$a_{ij} = a_{ij}^*$
$A = A^\dagger$	Hermitian	$a_{ij} = a_{ji}^*$
$A = (A^\dagger)^{-1}$	unitary	$\sum_k a_{ik} a_{jk}^* = \sum_k a_{ki}^* a_{kj} = \delta_{ij}$

Unitarity and hermiticity, as we will see, play central roles in this course: observables are Hermitian, changes of basis and time evolution are unitary.

Some famous (structured) matrices

- diagonal: $a_{ij} = 0$ for $i \neq j$
- upper/lower triangular: $a_{ij} = 0$ for $i > j$ / $i < j$
- upper/lower Hessenberg: $a_{ij} = 0$ for $i > j + 1$ / $i < j - 1$
- tridiagonal: $a_{ij} = 0$ for $|i - j| > 1$
- lower/upper banded with bandwidth p : $a_{ij} = 0$ for $i > j + p$ / $i < j - p$
- banded, block upper triangular, block lower triangular, ...

Many-body connection

Hamiltonian matrices in full configuration-interaction (shell-model) calculations are large, sparse and often banded. Exploiting sparsity is what makes them diagonalisable at all for systems with many single-particle states.

Pause to think: Hermitian, unitary, or both?

Pause to think #2

(about 3 min – discuss with your neighbour)

Consider the three matrices

$$\sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}.$$

Which of them are Hermitian? Which are unitary? Which are both? What does "both" imply for the eigenvalues?

Answer to pause #2

- σ_x : real symmetric, so Hermitian; $\sigma_x^\dagger \sigma_x = \sigma_x^2 = I$, so also unitary.
- H (Hadamard): real symmetric, Hermitian; $H^2 = I$, unitary. Both.
- S : $S^\dagger = \text{diag}(1, -i) \neq S$, not Hermitian; $S^\dagger S = I$, unitary.
- Hermitian $\Rightarrow$ real eigenvalues; unitary $\Rightarrow$ eigenvalues of modulus one. Both at once forces eigenvalues ± 1 and hence $A^2 = I$ (involutory). Check: σ_x and H both have eigenvalues ± 1 .

Defining basis states and quantum mechanical operators

We now carry the definitions of vectors and matrices over to quantum mechanics. A state vector with n components (representing some quantum mechanical degrees of freedom) is written in Dirac's **bra-ket** notation as

$$x = |x\rangle = \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_{n-1} \end{bmatrix} \quad (\text{ket, column vector}),$$
$$x^\dagger = \langle x| = [x_0^* \quad x_1^* \quad \cdots \quad x_{n-1}^*] \quad (\text{bra, row vector}).$$

The inner product of two vectors of the same length is

$$\langle y|x\rangle = \sum_{i=0}^{n-1} y_i^* x_i = y_0^* x_0 + y_1^* x_1 + \cdots + y_{n-1}^* x_{n-1},$$

and in particular $\langle x|x\rangle = \sum_i |x_i|^2$ is real and non-negative.

Note in passing that $|x\rangle^\dagger = \langle x|$, $\langle x|^\dagger = |x\rangle$ and $(|x\rangle^\dagger)^\dagger = |x\rangle$.

Worked examples

Let $|x\rangle = \begin{bmatrix} 1 - i \\ 2 + i \end{bmatrix}$. Then

$$\langle x|x \rangle = (1 + i)(1 - i) + (2 - i)(2 + i) = 2 + 5 = 7 \in \mathbb{R}, \quad \text{normalised: } |x\rangle \rightarrow \frac{1}{\sqrt{7}} \begin{bmatrix} 1 - i \\ 2 + i \end{bmatrix}.$$

Two different vectors,

$$|x\rangle = \begin{bmatrix} -1 \\ 2i \\ 1 \end{bmatrix}, \quad |y\rangle = \begin{bmatrix} 1 \\ 0 \\ i \end{bmatrix} : \quad \langle x|y \rangle = -1 + i \neq \langle y|x \rangle = -1 - i.$$

The rule

$$\langle x|y \rangle^* = \langle y|x \rangle.$$

Outer products in Dirac notation

The outer product $|x\rangle\langle y|$ is an $n \times n$ matrix with elements $x_i y_j^*$,

$$|x\rangle\langle y| = \begin{bmatrix} x_0 y_0^* & x_0 y_1^* & \cdots & x_0 y_{n-1}^* \\ x_1 y_0^* & x_1 y_1^* & \cdots & x_1 y_{n-1}^* \\ \vdots & \vdots & \ddots & \vdots \\ x_{n-1} y_0^* & x_{n-1} y_1^* & \cdots & x_{n-1} y_{n-1}^* \end{bmatrix}.$$

Assume a two-level system with states $|\phi_0\rangle$ and $|\phi_1\rangle$. They could represent effective degrees of freedom of a single particle (fermion or boson), or effective many-body degrees of freedom. In quantum computing one looks for systems whose low-lying states can serve as **computational basis states**; in many-body physics, because the degrees of freedom explode, we search for effective ways to reduce the dimensionality to something a given method can handle.

Computational basis states and projection operators

Relabel the two states as an orthonormal basis (ONB),

$$|\phi_0\rangle = |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |\phi_1\rangle = |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \langle 1|0\rangle = 0, \quad \langle 0|0\rangle = \langle 1|1\rangle = 1.$$

The identity is the sum of the outer products,

$$I = \sum_{i=0}^1 |i\rangle\langle i| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

and the two pieces are the **projection operators**

$$P = |0\rangle\langle 0| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad Q = |1\rangle\langle 1| = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

They are idempotent, $P^2 = P$, $Q^2 = Q$; their determinants vanish (so they are not unitary); they commute, $[P, Q] = 0$, and $PQ = 0$, $P + Q = I$.

Superposition, and acting with projection operators

A general state is a linear expansion in the computational basis,

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad \alpha = \langle 0|\psi\rangle, \quad \beta = \langle 1|\psi\rangle,$$

the coefficients being the overlaps with the basis states ("the state is in a superposition of $|0\rangle$ and $|1\rangle$ "). Its norm is

$$\langle\psi|\psi\rangle = |\alpha|^2\langle 0|0\rangle + |\beta|^2\langle 1|1\rangle = |\alpha|^2 + |\beta|^2 = 1.$$

Acting with the projectors,

$$P|\psi\rangle = |0\rangle\langle 0|(\alpha|0\rangle + \beta|1\rangle) = \alpha|0\rangle, \quad Q|\psi\rangle = \beta|1\rangle :$$

P *projects out* the $|0\rangle$ component with coefficient α and Q the $|1\rangle$ component with coefficient β . (Check it by multiplying the matrices with the vectors $|0\rangle$ and $|1\rangle$.)

Why projection operators matter for us

Many-body connection

In many-body theory the projectors $\hat{P}$ (onto occupied states, or onto a chosen model space) and $\hat{Q} = 1 - \hat{P}$ (onto everything else) define the particle-hole decomposition used in Hartree-Fock theory, perturbation theory and coupled cluster theory, and they define *effective* Hilbert spaces: the exact problem in the full space is replaced by an effective Hamiltonian acting in the $\hat{P}$ -space only.

Remember the properties: $\hat{P}^2 = \hat{P}$, $\hat{Q}^2 = \hat{Q}$, $\hat{P}\hat{Q} = \hat{Q}\hat{P} = 0$, $\hat{P} + \hat{Q} = 1$, and $\det \hat{P} = 0$ – a projector is not invertible, so it can never be a unitary transformation. Projection throws information away; unitary rotation never does.

Measurements

The probability of a measurement giving a certain result is the weight of the relevant basis state in the state vector; after the measurement the system is in the state associated with the result. For $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$:

1. A measurement yields only one of the states, either $|0\rangle$ or $|1\rangle$.
2. The probability of $|0\rangle$ is $\alpha^*\alpha = |\alpha|^2$; the probability of $|1\rangle$ is $\beta^*\beta = |\beta|^2$.
3. The outcomes sum to $\alpha^*\alpha + \beta^*\beta = 1$ because the state is normalised.

Density matrix of a pure state

$$\begin{aligned}\rho &= |\psi\rangle\langle\psi| = \alpha\alpha^*|0\rangle\langle 0| + \alpha\beta^*|0\rangle\langle 1| + \beta\alpha^*|1\rangle\langle 0| + \beta\beta^*|1\rangle\langle 1| \\ &= \begin{bmatrix} \alpha\alpha^* & \alpha\beta^* \\ \beta\alpha^* & \beta\beta^* \end{bmatrix}, \quad \text{tr } \rho = |\alpha|^2 + |\beta|^2 = 1.\end{aligned}$$

Density matrices generalise to mixed states and subsystems; they are central to quantum information theory and to reduced-density-matrix many-body methods.

Pauli matrices

The Pauli matrices, and other simple 2×2 matrices, play a central role: quantum gates, and rewriting creation and annihilation operators.

$$\sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \sigma_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad \sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

Properties (easy to verify):

- involutory: $\sigma_x^2 = \sigma_y^2 = \sigma_z^2 = \mathbb{I}$
- unitary: $\sigma_\alpha^\dagger = \sigma_\alpha^{-1}$
- Hermitian: $\sigma_\alpha^\dagger = \sigma_\alpha$
- $\det \sigma_\alpha = -1$ for each α
- $[\sigma_x, \sigma_y] = 2i\sigma_z$ and cyclic permutations

Action on the computational basis:

$$\begin{aligned} \sigma_x|0\rangle &= |1\rangle, & \sigma_x|1\rangle &= |0\rangle, \\ \sigma_z|0\rangle &= +|0\rangle, & \sigma_z|1\rangle &= -|1\rangle, \end{aligned}$$

and on a superposition

$$\sigma_x|\psi\rangle = \beta|0\rangle + \alpha|1\rangle, \quad \sigma_y|\psi\rangle = i\beta|0\rangle - i\alpha|1\rangle.$$

Pauli matrices: commutation relations and why we care

$$[\sigma_x, \sigma_y] = 2i\sigma_z, \quad [\sigma_y, \sigma_z] = 2i\sigma_x, \quad [\sigma_z, \sigma_x] = 2i\sigma_y, \quad \{\sigma_\alpha, \sigma_\beta\} = 2\delta_{\alpha\beta}I.$$

Quantum computing connection

The Pauli matrices are the single-qubit gates X, Y, Z ; with the Hadamard gate $H = (\sigma_x + \sigma_z)/\sqrt{2}$ and the phase gate T they form a universal single-qubit gate set.

Many-body connection

Via the Jordan-Wigner transformation, fermionic creation and annihilation operators become strings of Pauli matrices, $\sigma_\pm = (\sigma_x \pm i\sigma_y)/2$ raising and lowering a single level. That is how a fermionic Hamiltonian is written on a qubit register – and it is why the anticommutator above, not only the commutator, will matter.

Before we proceed with other matrices, let us define various basis sets and their notation: our *computational basis*.

Pause to think: what does σ_x do to the density matrix?

Pause to think #3

(about 3 min – discuss with your neighbour)

Take $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ with density matrix $\rho = |\psi\rangle\langle\psi|$.

1. Write down $|\psi'\rangle = \sigma_x|\psi\rangle$ and its density matrix ρ' .
2. How is ρ' related to ρ in matrix language? (Hint: $\sigma_x^\dagger = \sigma_x$.)
3. What are $\text{tr } \rho'$ and ρ'^2 ?

Answer to pause #3

1. $|\psi'\rangle = \beta|0\rangle + \alpha|1\rangle$, so $\rho' = \begin{bmatrix} \beta\beta^* & \beta\alpha^* \\ \alpha\beta^* & \alpha\alpha^* \end{bmatrix}$: the diagonal entries (the probabilities) have been swapped.
2. $\rho' = \sigma_x |\psi\rangle \langle \psi| \sigma_x^\dagger = \sigma_x \rho \sigma_x$. A unitary acts on a density matrix by conjugation, $\rho \rightarrow U \rho U^\dagger$.
3. $\text{tr } \rho' = \text{tr}(\sigma_x \rho \sigma_x) = \text{tr}(\rho \sigma_x^2) = \text{tr } \rho = 1$ (cyclic trace), and $\rho'^2 = \sigma_x \rho \sigma_x \sigma_x \rho \sigma_x = \sigma_x \rho^2 \sigma_x = \rho'$: still a pure state. Unitary evolution preserves trace and purity.

Thursday, lecture 2: Bases, unitary transformations, tensor products and Hamiltonians

Orthonormal bases and the completeness relation

A set of states $\{|\phi_i\rangle\}_{i=0}^{n-1}$ is an **orthonormal basis** (ONB) of an n -dimensional Hilbert space if

$$\langle\phi_j|\phi_i\rangle = \delta_{ij} \quad \text{and} \quad \sum_{i=0}^{n-1} |\phi_i\rangle\langle\phi_i| = \mathbb{I} \quad (\text{completeness, resolution of the identity}).$$

Any state can then be expanded as

$$|\psi\rangle = \sum_i \langle\phi_i|\psi\rangle |\phi_i\rangle,$$

with $\langle\phi_i|\psi\rangle$ the projection (overlap) of $|\psi\rangle$ onto $|\phi_i\rangle$. Inserting the identity $\sum_i |\phi_i\rangle\langle\phi_i|$ between operators is the single most used trick of the course.

Hamiltonians and basis functions

It is normal to choose a basis defined as the eigenfunctions of *parts* of the full Hamiltonian. The typical situation consists of the solutions of the one-body part,

$$\hat{h}_0|\phi_i\rangle = \epsilon_i|\phi_i\rangle,$$

the single-particle basis $\phi_i(\mathbf{r}) = \langle \mathbf{r} | \phi_i \rangle$ labelled by the quantum numbers i and the coordinate $\mathbf{r}$.

Examples: hydrogen-like functions (Coulomb attraction from the nucleus), harmonic oscillator functions in one, two or three dimensions, plane waves for infinite matter.

Many-body connection

This single-particle basis is the starting point for Hartree-Fock, MBPT, FCI, coupled cluster and quantum Monte Carlo. Although it contains infinitely many states, in practice we always truncate. Computationally, $\hat{h}_0\phi = \epsilon\phi$ is an eigenvalue problem: discretise on a grid or expand in a finite basis, and diagonalise. The oscillator functions on a grid are the basis used in all numerical examples this week (notebook).

Unitarity

Essentially all matrices we meet this week are unitary. This is essential in quantum mechanics: the evolution of a closed quantum system is described by unitary operations only.

Definition

U is unitary if $U^\dagger U = U U^\dagger = I$, that is $U^{-1} = U^\dagger$, with matrix elements $u_{ij}^\dagger = u_{ji}^*$.

A new state $|\psi_i\rangle$ defined as a linear expansion in an orthonormal basis $|\phi_j\rangle$,

$$|\psi_i\rangle = \sum_j u_{ij} |\phi_j\rangle,$$

is obtained by a unitary transformation if the coefficient matrix (u_{ij}) is unitary.

Because the Hermitian conjugate of a unitary matrix is its inverse, every unitary transformation is **reversible**: quantum mechanics is reversible with a probabilistic determinism, classical mechanics reversible in a deterministic way.

Unitary transformations preserve orthogonality

Consider an orthonormal basis of vectors v_i ,

$$v_i = \begin{bmatrix} v_{i1} \\ \vdots \\ v_{in} \end{bmatrix}, \quad v_j^T v_i = \delta_{ij}.$$

An orthogonal or unitary transformation $w_i = Uv_i$ preserves the dot product and the orthogonality since

$$w_j^T w_i = (Uv_j)^T Uv_i = v_j^T \underbrace{U^T U}_{=I} v_i = v_j^T v_i = \delta_{ij}.$$

In Dirac notation: if $|\psi_i\rangle = \sum_j u_{ij}|\phi_j\rangle$ with (u_{ij}) unitary, then $\langle\psi_j|\psi_i\rangle = \delta_{ij}$.

This property is extremely useful when we build up a basis of many-body determinant-based states: we can search over single-particle bases without ever having to re-orthogonalise.

Why are only unitary transformations allowed?

Let $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ be normalised and let $|\phi\rangle = \gamma|0\rangle + \delta|1\rangle = U|\psi\rangle$,

$$\begin{bmatrix} \gamma \\ \delta \end{bmatrix} = \begin{bmatrix} u_{00} & u_{01} \\ u_{10} & u_{11} \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix}, \quad \text{i.e.} \quad |\phi\rangle = U|\psi\rangle, \quad \langle\phi| = \langle\psi|U^\dagger.$$

The key lies in how the inner product transforms:

$$\langle\phi|\phi\rangle = \langle\psi|U^\dagger U|\psi\rangle = \langle\psi|\psi\rangle = 1!$$

Probabilities must sum to one, so $U^\dagger U = I$ is a physical requirement: unitary transformations are **rotations (and reflections) in state space** which preserve the length of the state vector.

Quantum computing connection

Every quantum gate is a unitary matrix. The evolution of a closed quantum system is a sequence of unitary operations – the quantum analogue of classical reversibility.

Pause to think: eigenvalues in a rotated basis

Pause to think #4

(about 4 min – discuss with your neighbour)

Let O be Hermitian with $O|\phi_\lambda\rangle = o_\lambda|\phi_\lambda\rangle$, and let U be unitary. Define the rotated states $|\psi_\lambda\rangle = U|\phi_\lambda\rangle$.

1. Which operator has $|\psi_\lambda\rangle$ as eigenvectors, and with which eigenvalues?
2. Is the expectation value $\langle\psi|O|\psi\rangle$ of a Hermitian operator always real? Why?

(This is exercise 1 of this week's exercise set.)

Answer to pause #4

1. $O' = UOU^\dagger$, since

$$O'|\psi_\lambda\rangle = UO\underbrace{U^\dagger U}_I|\phi_\lambda\rangle = o_\lambda U|\phi_\lambda\rangle = o_\lambda|\psi_\lambda\rangle.$$

The eigenvalues are unchanged – a unitary change of basis is a *similarity transformation*, and the spectrum is a property of the operator, not of the basis we write it in.

2. $\langle\psi|O|\psi\rangle^* = \langle\psi|O^\dagger|\psi\rangle = \langle\psi|O|\psi\rangle$, so yes. This is why observables are represented by Hermitian operators.

Tensor products

Consider two vectors of length $n = 2$, $|x\rangle = \begin{bmatrix} x_0 \\ x_1 \end{bmatrix}$ and $|y\rangle = \begin{bmatrix} y_0 \\ y_1 \end{bmatrix}$. Their tensor (Kronecker) product is the vector of length 4

$$|x\rangle \otimes |y\rangle = |xy\rangle = \begin{bmatrix} x_0 y_0 \\ x_0 y_1 \\ x_1 y_0 \\ x_1 y_1 \end{bmatrix}.$$

For the computational basis states:

$$|0\rangle \otimes |0\rangle = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = |00\rangle, \quad |0\rangle \otimes |1\rangle = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} = |01\rangle, \quad |1\rangle \otimes |0\rangle = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} = |10\rangle, \quad |1\rangle \otimes |1\rangle = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = |11\rangle.$$

These four states form an ONB for the two-qubit space $\mathbb{C}^4 = \mathbb{C}^2 \otimes \mathbb{C}^2$.

The exponential wall

The most general state of N qubits (or N spin- $\frac{1}{2}$ particles) is

$$|\Psi\rangle = \sum_{i_1 i_2 \dots i_N} \psi_{i_1 i_2 \dots i_N} |i_1\rangle \otimes |i_2\rangle \otimes \dots \otimes |i_N\rangle,$$

and needs 2^N complex amplitudes. For $N \approx 300$ this exceeds the number of atoms in the observable universe.

For fermions: the number of Slater determinants obtained by distributing N particles over n single-particle states is $\binom{n}{N}$; for $n = 40$, $N = 20$ that is already 1.4×10^{11} .

Many-body connection

The full N -particle Hilbert space is a tensor product of N single-particle spaces. FCI works in this space; HF, MBPT, coupled cluster, DMRG, ... are strategies to find the relevant low-dimensional part of it.

Quantum computing connection

A quantum computer with N qubits *inherently* represents a 2^N -dimensional state vector – the source of its potential power and the reason quantum simulation of many-body systems is a leading application.

Pause to think: how many numbers?

Pause to think #5

(about 3 min – discuss with your neighbour)

1. A state of N qubits needs 2^N complex amplitudes. With 16 bytes per complex number, how many qubits can you store in 16 GB of memory? And in the memory of the largest supercomputer you can think of (say 10 PB)?
2. Two qubits in the state $|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$: can you write it as a tensor product $|x\rangle \otimes |y\rangle$ of two single-qubit states?

Answer to pause #5

1. $16\text{ GB} = 2^{34}$ bytes $= 2^{30}$ complex numbers: $N = 30$ qubits. $10\text{ PB} \approx 2^{53}$ bytes gives $N \approx 49$. Doubling the memory buys you *one* more qubit. That is the exponential wall in practice.
2. No. A product state has amplitudes $\Psi_{ij} = x_i y_j$, a rank-one 2×2 matrix; here $\Psi = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ has rank two. The state is *entangled*. The same rank argument (the Schmidt decomposition) will later tell us how much a many-body state deviates from a product of single-particle states.

Spectral decomposition of Hermitian operators

A Hermitian operator $A = A^\dagger$ has real eigenvalues, orthogonal eigenvectors for distinct eigenvalues, and is diagonalisable. Nontrivial solutions of $A|\psi\rangle = \lambda|\psi\rangle$ require $\det(A - \lambda I) = 0$. With $\{|i\rangle\}$ the eigenbasis and $P_i = |i\rangle\langle i|$ the projectors, any $|\psi\rangle = \sum_i \alpha_i |i\rangle$ gives

$$A|\psi\rangle = \sum_i \alpha_i \lambda_i |i\rangle = \sum_i \lambda_i P_i |\psi\rangle \quad \Rightarrow \quad \boxed{A = \sum_{i=0}^{n-1} \lambda_i |i\rangle\langle i| = \sum_{i=0}^{n-1} \lambda_i P_i}$$

For a 2×2 operator with eigenstates $|\psi_a\rangle = \alpha_0|0\rangle + \alpha_1|1\rangle$, $|\psi_b\rangle = \beta_0|0\rangle + \beta_1|1\rangle$:

$$A = \lambda_a \begin{bmatrix} |\alpha_0|^2 & \alpha_0 \alpha_1^* \\ \alpha_1 \alpha_0^* & |\alpha_1|^2 \end{bmatrix} + \lambda_b \begin{bmatrix} |\beta_0|^2 & \beta_0 \beta_1^* \\ \beta_1 \beta_0^* & |\beta_1|^2 \end{bmatrix}.$$

Spectral decomposition: the two connections

Many-body connection

Every diagonalisation of a Hamiltonian matrix is a spectral decomposition: FCI diagonalises the N -body Hamiltonian in the basis of many-particle determinants, Hartree-Fock diagonalises the Fock matrix (a single-particle Hermitian operator) to find the optimal orbitals.

Quantum computing connection

Any unitary can be written as $U = e^{iH}$ for a Hermitian generator H , and the spectral decomposition of H gives $U = \sum_i e^{i\lambda_i} |i\rangle\langle i|$. This links the Hamiltonian to the time-evolution operator $e^{-i\hat{H}t/\hbar}$ and is the basis of the variational quantum eigensolver and the unitary coupled-cluster ansatz.

Functions of a Hermitian operator are defined through the spectral decomposition:
 $f(A) = \sum_i f(\lambda_i) |i\rangle\langle i|$. We will use e^A , $A^{-1/2}$ and $\ln A$ later in the course.

The Hamiltonian: definitions for the rest of the course

Throughout these lectures we assume that the interacting part of the Hamiltonian can be approximated by a **two-body interaction**. The Hamiltonian is the sum of a one-body part (kinetic energy and an eventual external field) and a two-body part,

$$\hat{H} = \hat{H}_0 + \hat{H}_I = \sum_{i=1}^N \hat{h}_0(x_i) + \sum_{i<j}^N \hat{v}(r_{ij}), \quad \hat{h}_0(x_i) = \hat{t}(x_i) + \hat{u}_{\text{ext}}(x_i),$$

with x_i the coordinates *and* spin of particle i . The one-body potential $\hat{u}_{\text{ext}}$ is normally a harmonic oscillator, or the Coulomb attraction from a nucleus; other choices are possible, for instance the self-consistent Hartree-Fock potential.

The single-particle functions ψ_α are eigenfunctions of $\hat{h}_0$,

$$\hat{h}_0(x_i)\psi_\alpha(x_i) = (\hat{t}(x_i) + \hat{u}_{\text{ext}}(x_i))\psi_\alpha(x_i) = \varepsilon_\alpha\psi_\alpha(x_i),$$

with ε_α the *non-interacting* (unperturbed) single-particle energies. Without $\hat{H}_I$ the total energy is the sum of the occupied ε_α . **Everything that makes many-body physics hard is the correction to that statement.**

Summary of Thursday, and what comes on Friday

Today

- Vectors, matrices, inner and outer products; bra-ket notation
- Projection operators P , Q and density matrices as outer products
- Pauli matrices: Hermitian, unitary, involutory
- ONB, completeness, unitary transformations preserve orthogonality and probabilities
- Tensor products and the 2^N wall; spectral decomposition
- $\hat{H} = \hat{H}_0 + \hat{H}_I$ and the single-particle basis
 $\hat{h}_0\psi_\alpha = \varepsilon_\alpha\psi_\alpha$

Friday

- Identical particles: permutation symmetry, bosons and fermions
- Determinants and the Slater determinant
- Changing the basis: $|C||\Phi|$
- The variational principle
- Expectation values of $\hat{H}_0$ and $\hat{H}_I$: direct and exchange terms
- The energy as a functional of the coefficients – the road to Hartree-Fock
- Exercise session

Before Friday: run the first cells of `week34.ipynb` (inner/outer products, projectors, Pauli matrices, unitary rotations) and look at exercise 1 of the week-34 set.

Friday, lecture 3: Fermions, bosons and Slater determinants

Where we are

Thursday gave us a language: states are vectors, operators are (Hermitian or unitary) matrices, bases are changed by unitary transformations, and the N -particle space is a tensor product.

Today we put it to work. We shall see that

- the central object of fermionic many-body theory, the **Slater determinant**, is a determinant in exactly the usual sense;
- changing the single-particle basis is a unitary transformation, and the determinant changes only by a phase;
- the expectation value of $\hat{H}$ in a Slater determinant is a simple sum of one- and two-body matrix elements, the latter with a **direct** and an **exchange** part;
- minimising that expectation value is the Hartree-Fock method – an eigenvalue problem again.

Nothing today is new mathematics. What is new is the physics the mathematics is asked to carry.

The Hamiltonian is invariant under permutations

Recall $\hat{H} = \sum_{i=1}^N \hat{h}_0(x_i) + \sum_{i < j}^N \hat{v}(r_{ij})$: it is symmetric under the interchange of any two particles. Let $\hat{P}_{ij}$ be the operator that interchanges particles i and j . Then

$$[\hat{H}, \hat{P}_{ij}] = 0,$$

so the eigenfunctions Ψ_λ of $\hat{H}$ can be chosen to be eigenfunctions of $\hat{P}_{ij}$ as well,

$$\hat{P}_{ij} \Psi_\lambda(x_1, \dots, x_i, \dots, x_j, \dots, x_N) = \beta \Psi_\lambda(x_1, \dots, x_i, \dots, x_j, \dots, x_N).$$

Applying the permutation twice returns the original function, so $\beta^2 = 1$ and $\beta = \pm 1$:

Bosons and fermions

- Bosons: $\beta = +1$, the wave function is symmetric under every transposition.
- Fermions: $\beta = -1$, the wave function is antisymmetric (the **Pauli principle**). Fermions are our main concern in this course.

Pause to think: why only ± 1 , and what about three particles?

Pause to think #6

(about 3 min – discuss with your neighbour)

1. We argued $\beta = \pm 1$ from $\hat{P}_{ij}^2 = 1$. Which additional property of $\hat{P}_{ij}$ guarantees that β is a *phase* with $|\beta| = 1$ in the first place?
2. For three identical fermions, what is the sign picked up under the cyclic permutation $(x_1, x_2, x_3) \rightarrow (x_2, x_3, x_1)$?

Answer to pause #6

1. $\hat{P}_{ij}$ is unitary (it preserves the norm $\int |\Psi|^2$), so its eigenvalues have modulus one; combined with $\hat{P}_{ij}^2 = 1$ this leaves exactly ± 1 . (It is also Hermitian, $\hat{P}_{ij}^\dagger = \hat{P}_{ij}^{-1} = \hat{P}_{ij}$, so the eigenvalues are real – the same "Hermitian and unitary" situation as for the Pauli matrices yesterday.)
2. A cyclic permutation of three objects is a product of *two* transpositions, e.g. swap $1 \leftrightarrow 2$ then $1 \leftrightarrow 3$, so the sign is $(-1)^2 = +1$. Even permutations give $+$, odd ones $-$. The parity of a permutation is exactly what enters the definition of a determinant – next slide.

Determinant algebra

Since we mainly deal with fermions (identical, indistinguishable particles) we form an ansatz for a state in terms of Slater determinants built from a chosen basis of single-particle functions. First, determinants themselves. For an $n \times n$ matrix A ,

$$\det(A) = |A| = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} = \sum_{i=1}^{n!} (-1)^{p_i} \hat{P}_i a_{11} a_{22} \cdots a_{nn},$$

where $\hat{P}_i$ permutes the column indices $1, 2, \dots, n$, the sum runs over all $n!$ permutations, and p_i is the number of transpositions needed to bring a permutation back to the natural ordering $a_{11} a_{22} \cdots a_{nn}$.

The simplest case, $n = 2$:

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = (-1)^0 a_{11} a_{22} + (-1)^1 a_{12} a_{21},$$

the second term obtained by interchanging the column indices 1 and 2.

Two determinant properties we shall need

Linearity in a column.

$$\begin{vmatrix} \alpha_1 b_{11} + \alpha_2 b_{12} & a_{12} \\ \alpha_1 b_{21} + \alpha_2 b_{22} & a_{22} \end{vmatrix} = \alpha_1 \begin{vmatrix} b_{11} & a_{12} \\ b_{21} & a_{22} \end{vmatrix} + \alpha_2 \begin{vmatrix} b_{12} & a_{12} \\ b_{22} & a_{22} \end{vmatrix},$$

and in general

$$\begin{vmatrix} a_{11} & \cdots & \sum_{k=1}^n c_k b_{1k} & \cdots & a_{1n} \\ \vdots & & \vdots & & \vdots \\ a_{n1} & \cdots & \sum_{k=1}^n c_k b_{nk} & \cdots & a_{nn} \end{vmatrix} = \sum_{k=1}^n c_k \begin{vmatrix} a_{11} & \cdots & b_{1k} & \cdots & a_{1n} \\ \vdots & & \vdots & & \vdots \\ a_{n1} & \cdots & b_{nk} & \cdots & a_{nn} \end{vmatrix}.$$

The product rule. If every element is a linear combination, $a_{ij} = \sum_{k=1}^n b_{ik} c_{kj}$, i.e. $A = BC$, then

$$\begin{vmatrix} \sum_k b_{1k} c_{k1} & \cdots & \sum_k b_{1k} c_{kn} \\ \vdots & & \vdots \\ \sum_k b_{nk} c_{k1} & \cdots & \sum_k b_{nk} c_{kn} \end{vmatrix} = \det(B) \det(C).$$

This is the single most important algebraic fact this week; convince yourself for $n = 2$ (exercise 2).

A warning: definition is not algorithm

Evaluating a determinant from the permutation sum costs $\mathcal{O}(n!)$ operations. The LU decomposition gives the same number in $\mathcal{O}(n^3)$ as the product of the diagonal of U.

n	from the definition, $n!$	from LU, $\sim \frac{2}{3}n^3$
5	1.2×10^2	8.3×10^1
10	3.6×10^6	6.7×10^2
20	2.4×10^{18}	5.3×10^3
50	3.0×10^{64}	8.3×10^4

Every Slater determinant evaluated in a Monte Carlo or configuration-interaction code is computed by LU (or updated cleverly), never from the permutation sum.

```
import numpy as np
M = np.random.default_rng(1).normal(size=(6, 6))
print(np.linalg.det(M)) # LU under the hood,  $\mathcal{O}(n^3)$ 
```

The Slater determinant

Given N single-particle orbitals $\psi_\alpha, \psi_\beta, \dots, \psi_\sigma$, the Slater determinant is

$$\Phi(x_1, x_2, \dots, x_N; \alpha, \beta, \dots, \sigma) = \frac{1}{\sqrt{N!}} \begin{vmatrix} \psi_\alpha(x_1) & \psi_\alpha(x_2) & \cdots & \psi_\alpha(x_N) \\ \psi_\beta(x_1) & \psi_\beta(x_2) & \cdots & \psi_\beta(x_N) \\ \vdots & \vdots & \ddots & \vdots \\ \psi_\sigma(x_1) & \psi_\sigma(x_2) & \cdots & \psi_\sigma(x_N) \end{vmatrix},$$

where x_i stands for the coordinates and spin of particle i and $\alpha, \beta, \dots, \sigma$ are the quantum numbers of the occupied orbitals.

Two physical properties from two properties of determinants

- Interchanging two particles interchanges two *columns*: the sign flips – the wave function is antisymmetric.
- Two particles in the same orbital give two identical *rows*: the determinant vanishes – no two fermions in the same single-particle state.

The Pauli principle is not an extra postulate; it is a property of determinants.

Hartree function and the antisymmetrisation operator

Write the unsymmetrised product of single-particle functions, the **Hartree function**,

$$\Phi_H(x_1, x_2, \dots, x_N; \alpha, \beta, \dots, \nu) = \psi_\alpha(x_1)\psi_\beta(x_2)\cdots\psi_\nu(x_N).$$

The Slater determinant is the same as

$$\Phi = \frac{1}{\sqrt{N!}} \sum_P (-)^P \hat{P} \Phi_H = \sqrt{N!} \hat{A} \Phi_H, \quad \hat{A} = \frac{1}{N!} \sum_P (-)^P \hat{P},$$

with $\hat{A}$ the antisymmetrisation operator, the sum over all $N!$ permutations and p the parity. Two properties do all the work in the next lecture:

$$[\hat{H}_0, \hat{A}] = [\hat{H}_I, \hat{A}] = 0 \quad \text{and} \quad \hat{A}^2 = \hat{A},$$

the first because $\hat{H}_0$ and $\hat{H}_I$ are invariant under all permutations, the second because permuting an already antisymmetrised function reproduces it up to the sign that the prefactor removes. $\hat{A}$ is a *projector* onto the antisymmetric subspace – the same idea as $P = |0\rangle\langle 0|$ yesterday.

A Slater determinant is an uncorrelated state

Many-body connection

The Hartree function is a tensor product $|\alpha\rangle \otimes |\beta\rangle \otimes \cdots \otimes |\nu\rangle$ of the kind introduced yesterday; the Slater determinant is its antisymmetric projection. In this precise sense a single determinant is an *uncorrelated* state: antisymmetrisation introduces exactly the correlation demanded by statistics and no more.

Made quantitative: the one-body density matrix $\rho_{\lambda\mu} = \sum_i C_{i\lambda}^* C_{i\mu}$ of a Slater determinant is a projector, and its eigenvalues (the *occupation numbers*) are exactly 1 for the N occupied orbitals and 0 for all others. Any deviation from 0 and 1 measures genuine, dynamical correlation.

In the notebook: for three particles in the lowest oscillator orbitals the occupation numbers come out as $(1, 1, 1, 0, 0, 0, 0)$ to machine precision.

Pause to think: two fermions in two orbitals

Pause to think #7

(about 4 min – discuss with your neighbour)

Two particles, two orthonormal orbitals ψ_a and ψ_b .

1. Write out the 2×2 Slater determinant $\Phi(x_1, x_2)$ explicitly.
2. Show by direct substitution that $\Phi(x_2, x_1) = -\Phi(x_1, x_2)$ and that Φ vanishes identically if $b = a$.
3. Check that Φ is normalised if ψ_a, ψ_b are orthonormal. Where did the $1/\sqrt{N!}$ come from?

Answer to pause #7

1. $\Phi(x_1, x_2) = \frac{1}{\sqrt{2}} [\psi_a(x_1)\psi_b(x_2) - \psi_b(x_1)\psi_a(x_2)]$.
2. Swapping $x_1 \leftrightarrow x_2$ swaps the two terms, hence the sign; setting $b = a$ makes the two terms identical and they cancel.
3. $\int |\Phi|^2 = \frac{1}{2} [\langle a|a\rangle\langle b|b\rangle + \langle b|b\rangle\langle a|a\rangle - \langle a|b\rangle\langle b|a\rangle - \langle b|a\rangle\langle a|b\rangle] = \frac{1}{2}[1+1-0-0] = 1$.
In general the $N!$ permutations each contribute one unit to the norm of the unnormalised sum, and the cross terms vanish by orthogonality – hence $1/\sqrt{N!}$.

The same computation done numerically (three particles, oscillator orbitals) is in the notebook: $\Phi(x_1, x_2, x_3) + \Phi(x_2, x_1, x_3)$ is zero to machine precision, and a repeated orbital gives $\Phi \sim 10^{-18}$.

Single-particle bases and unitary transformations

In practice one rarely knows the right orbitals in advance. The standard strategy is to expand them in a fixed, known orthonormal basis $\{\phi_\lambda\}$ – harmonic oscillator, hydrogen-like, plane waves, Gaussians – and to **vary the coefficients**,

$$\psi_p(x) = \sum_{\lambda} C_{p\lambda} \phi_{\lambda}(x), \quad \text{in Dirac notation} \quad |p\rangle = \sum_{\lambda} C_{p\lambda} |\lambda\rangle, \quad \phi_{\lambda}(r) = \langle r|\lambda\rangle.$$

This is the normal approach in Hartree-Fock theory. If the basis has infinitely many members the sum must be truncated – the first, and often the largest, approximation of a many-body calculation.

Two questions arise at once, and Thursday answered both:

1. Does the new basis remain orthonormal? Yes, *if and only if* C is unitary:

$$\langle p|q\rangle = \sum_{\lambda\mu} C_{p\lambda}^* C_{q\mu} \langle \lambda|\mu\rangle = \sum_{\lambda} C_{p\lambda}^* C_{q\lambda} = (CC^\dagger)_{qp} = \delta_{pq}.$$

2. What happens to the Slater determinant? Next slide.

The Slater determinant in the new basis

Substituting $\psi_p = \sum_{\lambda} C_{p\lambda} \phi_{\lambda}$ into the determinant, every entry becomes a sum:

$$\frac{1}{\sqrt{N!}} \begin{vmatrix} \psi_p(x_1) & \cdots & \psi_p(x_N) \\ \psi_q(x_1) & \cdots & \psi_q(x_N) \\ \vdots & \ddots & \vdots \\ \psi_t(x_1) & \cdots & \psi_t(x_N) \end{vmatrix} = \frac{1}{\sqrt{N!}} \begin{vmatrix} \sum_{\lambda} C_{p\lambda} \phi_{\lambda}(x_1) & \cdots & \sum_{\lambda} C_{p\lambda} \phi_{\lambda}(x_N) \\ \sum_{\lambda} C_{q\lambda} \phi_{\lambda}(x_1) & \cdots & \sum_{\lambda} C_{q\lambda} \phi_{\lambda}(x_N) \\ \vdots & \ddots & \vdots \\ \sum_{\lambda} C_{t\lambda} \phi_{\lambda}(x_1) & \cdots & \sum_{\lambda} C_{t\lambda} \phi_{\lambda}(x_N) \end{vmatrix}.$$

The matrix of orbitals is $M = C\Phi$ with $\Phi_{\lambda j} = \phi_{\lambda}(x_j)$, so by the product rule

$$\det(M) = \det(C) \det(\Phi)$$

Only a phase

If C is unitary then

$$\det(C)^* \det(C) = \det(C^\dagger) \det(C) = \det(C^\dagger C) = \det(I) = 1, \quad \text{so} \quad \det(C) = e^{i\theta},$$

and the rotated Slater determinant differs from the original **only by a phase** $e^{i\theta}$. Since all physical predictions depend on $|\Psi|^2$ or on expectation values, in which the phase cancels, it describes the *same* physical state.

Many-body connection

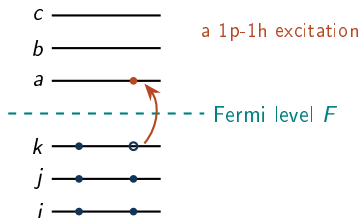
One may rotate the occupied orbitals among themselves at will without changing the many-body state, the density or the energy. This freedom is what makes canonical and localised Hartree-Fock orbitals equally valid descriptions of the same determinant, and it is the content of Thouless' theorem. In configuration interaction and coupled cluster theory it is why the choice of reference orbitals is a matter of convenience rather than principle.

Numerically (notebook): a random orthogonal rotation of three occupied orbitals changes the energy only at the level of machine precision ($\lesssim 10^{-15}$); a rotation mixing occupied and unoccupied orbitals changes it by more than a factor of two.

Particle-hole notation and the Fermi level

Given a reference Slater determinant – usually the N lowest single-particle levels filled – the single-particle states split into occupied and unoccupied. The dividing line is the **Fermi level** F . We write

$$\begin{aligned} i, j, k, l, \dots &\leq F && \text{occupied states, or } \textit{hole} \text{ states,} \\ a, b, c, d, \dots &> F && \text{unoccupied states, or } \textit{particle} \text{ states,} \\ p, q, r, s, \dots &&& \text{general single-particle states.} \end{aligned}$$



The reference determinant is treated as a new vacuum: exciting a particle from i to a creates a *particle* in a and leaves a *hole* in i . Second quantization (coming weeks) turns this into the particle-hole formalism; for now it is a convention worth adopting at once.

Friday, lecture 4: Expectation values and the energy functional

The variational principle

Let E_0 be the exact ground-state energy of $\hat{H}$. For any normalised trial function Φ ,

$$E_0 \leq E[\Phi] = \int \Phi^* \hat{H} \Phi \, d\tau, \quad \int \Phi^* \Phi \, d\tau = 1, \quad d\tau = dx_1 dx_2 \cdots dx_N.$$

Proof – the spectral decomposition and nothing more. Expand $|\Phi\rangle = \sum_k c_k |\Psi_k\rangle$ in the exact eigenstates $\hat{H}|\Psi_k\rangle = E_k |\Psi_k\rangle$, $\sum_k |c_k|^2 = 1$:

$$E[\Phi] = \sum_k |c_k|^2 E_k \geq E_0 \sum_k |c_k|^2 = E_0,$$

with equality only when Φ is the ground state itself.

This one inequality is the engine of most of the course

Trial function = one Slater determinant, minimise $\Rightarrow$ Hartree-Fock. Linear combination of determinants $\Rightarrow$ configuration interaction. Parametrised quantum circuit $\Rightarrow$ the variational quantum eigensolver. Correlated trial function sampled stochastically $\Rightarrow$ variational Monte Carlo.

Expectation values with a Slater determinant: the one-body part

We compute $E[\Phi]$ for $\Phi = \sqrt{N!} \hat{A} \Phi_H$, using only $[\hat{H}_0, \hat{A}] = 0$, $\hat{A}^2 = \hat{A}$ and the orthonormality of the orbitals.

$$\begin{aligned} \int \Phi^* \hat{H}_0 \Phi \, d\tau &= N! \int \Phi_H^* \hat{A} \hat{H}_0 \hat{A} \Phi_H \, d\tau && \text{(definition of } \Phi) \\ &= N! \int \Phi_H^* \hat{H}_0 \hat{A} \Phi_H \, d\tau && (\hat{A} \text{ commutes with } \hat{H}_0, \hat{A}^2 = \hat{A}) \\ &= \sum_{i=1}^N \sum_p (-)^p \int \Phi_H^* \hat{h}_0(x_i) \hat{P} \Phi_H \, d\tau && (\hat{A} \text{ by its definition, cancels } N!) \end{aligned}$$

The key step. The integrand factorises into single-particle integrals and $\hat{h}_0(x_i)$ touches only one coordinate. If $\hat{P}$ permutes any two particles, at least one remaining factor is an overlap $\langle \mu | \nu \rangle$ between two *different* orbitals, which vanishes. Only the identity permutation survives:

$$\int \Phi^* \hat{H}_0 \Phi \, d\tau = \sum_{i=1}^N \int \Phi_H^* \hat{h}_0(x_i) \Phi_H \, d\tau = \sum_{\mu=1}^N \int \psi_{\mu}^*(x) \hat{h}_0 \psi_{\mu}(x) \, dx \equiv \sum_{\mu=1}^N \langle \mu | \hat{h}_0 | \mu \rangle,$$

after integrating out the $N - 1$ spectator coordinates (each gives unity). The one-body energy is the sum of the diagonal matrix elements over the occupied orbitals.

The two-body part

The same three steps,

$$\int \Phi^* \hat{H}_I \Phi \, d\tau = N! \int \Phi_H^* \hat{A} \hat{H}_I \hat{A} \Phi_H \, d\tau = \sum_{i < j}^N \sum_p (-)^p \int \Phi_H^* \hat{v}(r_{ij}) \hat{P} \Phi_H \, d\tau,$$

but a different conclusion: because $\hat{v}(r_{ij})$ depends on two coordinates, the permutation that exchanges precisely those two particles no longer produces a vanishing overlap. **Two permutations survive:** the identity and the transposition $\hat{P}_{ij}$,

$$\int \Phi^* \hat{H}_I \Phi \, d\tau = \sum_{i < j}^N \int \Phi_H^* \hat{v}(r_{ij}) (1 - \hat{P}_{ij}) \Phi_H \, d\tau.$$

Writing out the two terms and integrating over the spectators,

$$\begin{aligned} \int \Phi^* \hat{H}_I \Phi \, d\tau = \frac{1}{2} \sum_{\mu=1}^N \sum_{\nu=1}^N & \left[\int \psi_{\mu}^*(x_i) \psi_{\nu}^*(x_j) \hat{v}(r_{ij}) \psi_{\mu}(x_i) \psi_{\nu}(x_j) \, dx_i dx_j \right. \\ & \left. - \int \psi_{\mu}^*(x_i) \psi_{\nu}^*(x_j) \hat{v}(r_{ij}) \psi_{\nu}(x_i) \psi_{\mu}(x_j) \, dx_i dx_j \right], \end{aligned}$$

the factor $\frac{1}{2}$ because the unrestricted double sum counts every pair twice.

Direct and exchange terms; notation for the matrix elements

- The first term is the **direct** or **Hartree** term: the classical electrostatic energy of one charge distribution in the field of another.
- The second is the **exchange** or **Fock** term. It has no classical counterpart; it exists solely because the wave function is antisymmetric – it is the trace left in the energy by the transposition $\hat{P}_{ij}$.

With $\psi_\alpha(x) = \langle x|\alpha\rangle$ we abbreviate

$$\langle \mu\nu|\hat{v}|\mu\nu\rangle = \int \psi_\mu^*(x_i)\psi_\nu^*(x_j)\hat{v}(r_{ij})\psi_\mu(x_i)\psi_\nu(x_j) dx_i dx_j,$$

$$\langle \mu\nu|\hat{v}|\nu\mu\rangle = \int \psi_\mu^*(x_i)\psi_\nu^*(x_j)\hat{v}(r_{ij})\psi_\nu(x_i)\psi_\mu(x_j) dx_i dx_j,$$

and define the **antisymmetrised** matrix element

$$\langle \mu\nu|\hat{v}|\mu\nu\rangle_{\text{AS}} = \langle \mu\nu|\hat{v}|\mu\nu\rangle - \langle \mu\nu|\hat{v}|\nu\mu\rangle.$$

Note $\langle \mu\mu|\hat{v}|\mu\mu\rangle_{\text{AS}} = 0$: a particle does not interact with itself. The self-interaction of the Hartree term is cancelled exactly by the exchange term – one reason Hartree-Fock is better behaved than the Hartree theory that preceded it.

Pause to think: which permutations survive?

Pause to think #8

(about 4 min – discuss with your neighbour)

Suppose the Hamiltonian also contained a *three-body* force $\sum_{i < j < k} \hat{w}(x_i, x_j, x_k)$.

1. How many permutations survive in the expectation value of $\hat{w}$ in a Slater determinant, and which are they?
2. Each surviving permutation comes with a sign. What is the sign pattern, and how many matrix elements $\langle \mu\nu\lambda | \hat{w} | \dots \rangle$ does the antisymmetrised three-body element contain?

Answer to pause #8

1. The operator touches three coordinates, so any permutation of those three particles leaves no vanishing spectator overlap: all $3! = 6$ permutations of $\{i, j, k\}$ survive (identity, three transpositions, two cyclic permutations), and nothing else.
2. Signs $+$ for the identity and the two cyclic (even) permutations, $-$ for the three transpositions: $\langle \mu\nu\lambda | \hat{w} | \mu\nu\lambda \rangle_{AS} = \langle \mu\nu\lambda | \hat{w} | \mu\nu\lambda \rangle - \langle \mu\nu\lambda | \hat{w} | \nu\mu\lambda \rangle - \langle \mu\nu\lambda | \hat{w} | \lambda\nu\mu \rangle - \langle \mu\nu\lambda | \hat{w} | \mu\lambda\nu \rangle + \langle \mu\nu\lambda | \hat{w} | \nu\lambda\mu \rangle + \langle \mu\nu\lambda | \hat{w} | \lambda\mu\nu \rangle$, six terms, with a prefactor $1/3!$ in front of the unrestricted triple sum.

The pattern – one term per permutation of the coordinates the operator touches, with the parity as sign – is exactly what Wick's theorem will automate in second quantization.

The energy functional

Collecting the one- and two-body results, the expectation value of the Hamiltonian in a Slater determinant is

$$E[\Phi] = \sum_{\mu=1}^N \langle \mu | \hat{h}_0 | \mu \rangle + \frac{1}{2} \sum_{\mu=1}^N \sum_{\nu=1}^N \langle \mu\nu | \hat{v} | \mu\nu \rangle_{AS},$$

or, in particle-hole notation with i, j running over the occupied states,

$$E[\Phi] = \sum_{i=1}^N \langle i | \hat{h}_0 | i \rangle + \frac{1}{2} \sum_{i,j=1}^N \langle ij | \hat{v} | ij \rangle_{AS}.$$

This is a functional of the occupied orbitals. To turn it into something we can *minimise*, insert the expansion $\psi_i = \sum_{\alpha} C_{i\alpha} \phi_{\alpha}$ in the fixed basis and let the coefficients carry the variation:

$$E[\Psi] = \sum_{i=1}^N \sum_{\alpha\beta} C_{i\alpha}^* C_{i\beta} \langle \alpha | \hat{h}_0 | \beta \rangle + \frac{1}{2} \sum_{i,j=1}^N \sum_{\alpha\beta\gamma\delta} C_{i\alpha}^* C_{j\beta}^* C_{i\gamma} C_{j\delta} \langle \alpha\beta | \hat{v} | \gamma\delta \rangle_{AS}.$$

Greek indices run over the full fixed basis, Latin indices over the N occupied orbitals only. Everything except the C 's is computed once and stored – and that is n^4 two-body numbers for a basis of size n .

Preview: minimising the functional is an eigenvalue problem

Minimising $E[\Psi]$ with respect to $C_{i\alpha}^*$ subject to orthonormality (derivation in a few weeks) gives the Hartree-Fock equations. Define the density matrix and the Fock matrix

$$\rho_{\gamma\delta} = \sum_{i=1}^N C_{i\gamma}^* C_{i\delta}, \quad f_{\alpha\beta} = \langle \alpha | \hat{h}_0 | \beta \rangle + \sum_{\gamma\delta} \rho_{\gamma\delta} \langle \alpha\gamma | \hat{v} | \beta\delta \rangle_{AS}.$$

The stationarity condition is

$$\sum_{\beta} f_{\alpha\beta} C_{i\beta} = \varepsilon_i C_{i\alpha},$$

a Hermitian eigenvalue problem – with the twist that f depends, through ρ , on the eigenvectors we are looking for. Guess, build f , diagonalise, rebuild, repeat: the **self-consistent field**.

```
def fock_matrix(C, h, v_as): # rows of C = occupied orbitals
    rho = C.T @ C # density matrix
    return h + np.einsum("cd,acbd->ab", rho, v_as)

for it in range(200): # the SCF loop
    eps, vecs = np.linalg.eigh(fock_matrix(C, h, v_as))
    C = vecs[:, :N].T # lowest N eigenvectors
```

Numbers: spinless fermions in a one-dimensional trap

Eight harmonic oscillator orbitals, softened Coulomb repulsion $v(x, y) = 1/\sqrt{(x - y)^2 + 0.25}$ (the model in the notebook):

N	E (lowest N orbitals)	E (Hartree-Fock)	gain
2	2.69018342	2.66308878	-2.7×10^{-2}
3	6.46801021	6.39016133	-7.8×10^{-2}
4	11.77084712	11.62305385	-1.5×10^{-1}

- The self-consistent orbitals always lie below the naive determinant built from the N lowest oscillator states (nine SCF iterations).
- For $N = 2$ the exact ground state in the same basis is obtained by diagonalising in the space of antisymmetric pairs. The difference $E_{\text{corr}} = E_{\text{exact}} - E_{\text{HF}} = -3.86 \times 10^{-3}$ defines the **correlation energy**; the largest amplitude of the exact state, 0.9930, sits on the Hartree-Fock configuration.
- Recovering the remaining 0.7 % of the wave function – a small residue of a large number – is what every method in the rest of the course is built to do.

Pause to think: what does the variational principle promise – and not?

Pause to think #9

(about 3 min – discuss with your neighbour)

1. The table shows $E_{\text{HF}} < E(\text{lowest orbitals})$. Does the variational principle guarantee that the Hartree-Fock energy is *above* the exact ground-state energy in the same basis? Below the exact energy in the infinite basis?
2. Can the correlation energy $E_{\text{exact}} - E_{\text{HF}}$ ever be positive?

Answer to pause #9

1. Within the truncated basis, yes: a single determinant is an allowed trial state of the eight-orbital problem, so $E_{\text{HF}} \geq E_{\text{exact}}^{(8)}$. Also $E_{\text{exact}}^{(8)} \geq E_0$ of the untruncated problem (the truncated space is a subspace of the full one), so E_{HF} is an upper bound to both. Nothing forces E_{HF} to be *close* to either.
2. No – by the same inequality, $E_{\text{corr}} \leq 0$ always, with equality only if the exact state is a single determinant (as at $g = 0$ in the notebook's interaction scan). Methods that are *not* variational (perturbation theory, coupled cluster) can overshoot; that is a recurring theme later.

Exercise session: week 34

The exercises are mainly reminders of the linear algebra elements we shall use throughout the course (full text in the exercise set for week 34).

Exercise 1: unitary transformations and orthogonality

U unitary, $|\psi_p\rangle = U|\phi_\lambda\rangle$ with $O|\phi_\lambda\rangle = o_\lambda|\phi_\lambda\rangle$, O Hermitian. a) Properties of unitary and orthogonal matrices. b) Properties of Hermitian matrices. c) Show that the new basis is orthonormal. d) Show that the eigenvalues in the new basis are unchanged.

Exercise 2: determinants

a) Use linearity in a column to show $\det(\sum_k b_{ik} c_{kj}) = \det(C) \det(B)$. b) Show that the Slater determinant in the basis $\psi_p = \sum_\lambda C_{p\lambda} \phi_\lambda$ equals $\det(C) \det(\Phi)$. c) Show that the new determinant differs from $\det(\Phi)$ by a complex phase only.

Hints: 1c) is the computation on the slide "Single-particle bases and unitary transformations"; 1d) is pause #5; 2a) is the $n^n \rightarrow n!$ counting argument (terms with two equal columns vanish); 2c) is $|\det C| = 1$.

What to do with the notebook

WeeklySlides/week34.ipynb contains, in this order:

1. Vectors, inner and outer products, bra-ket, the worked examples with $|x\rangle = (1 - i, 2 + i)^T$.
2. Projection operators, density matrix, Pauli matrices and their algebra (checks of all properties listed Thursday).
3. Unitary rotations: orthogonality preserved, norm preserved, eigenvalues unchanged under $UOU^\dagger$.
4. Tensor products, the four two-qubit states, the rank test for entanglement, and the 2^N memory count.
5. Determinants: permutation sum versus `numpy.linalg.det`, timing, linearity and the product rule checked numerically.
6. The harmonic oscillator basis on a grid and the Slater determinant: antisymmetry, Pauli principle, occupation numbers.
7. $\det(C)\det(\Phi)$: energy invariant under occupied-occupied rotations, not under occupied-virtual mixing.
8. The energy functional, a minimal SCF loop, and the correlation energy for two particles.

Play with it: change the interaction strength, the number of orbitals, the number of particles.

Summary of week 34

The mathematics

- States are vectors, observables Hermitian matrices, changes of basis unitary matrices.
- Projection operators and density matrices are outer products.
- The N -particle space is a tensor product: 2^N or $\binom{n}{N}$ amplitudes.
- Determinants are linear in each column and $\det(BC) = \det B \det C$.

The physics

- $[\hat{H}, \hat{P}_{ij}] = 0$ gives bosons (+) and fermions (-).
- The Slater determinant builds in antisymmetry and the Pauli principle.
- A unitary rotation of the occupied orbitals changes the determinant by a phase only.
- $E[\Phi] = \sum_i \langle i | \hat{h}_0 | i \rangle + \frac{1}{2} \sum_{ij} \langle ij | \hat{v} | ij \rangle_{AS}$: direct minus exchange.
- Minimising it is a self-consistent eigenvalue problem: Hartree-Fock.

Next week, and a closing remark

Next week (week 35)

Second quantization: creation and annihilation operators, anticommutation rules, the Hamiltonian in second quantized form, and the first steps towards Wick's theorem. Reading: chapter 3 of the lecture notes, Szabo and Ostlund chapter 2.

Everything we did this week by hand – moving $\hat{A}$ through $\hat{H}$, counting surviving permutations, keeping track of signs – second quantization does automatically. That is its entire purpose, and it is why the rest of the course is written in that language. Keep this week's derivations in mind: they are what the operator algebra is encoding.